

PRITOM PAUL

+880 1671 011145 | pritompaul98official@gmail.com | LinkedIn | GitHub | Dhaka, Bangladesh

SUMMARY

Software Engineer with 2+ years of experience in backend systems, industrial automation, and computer vision pipelines. I design and deploy solutions driven by data, with a focus on efficient and reliable systems across real world environments. I am now transitioning toward research, aiming to explore and contribute to learning based methods in applied settings.

RESEARCH INTERESTS

Computer Vision, Image Processing, Deep Learning, Machine Learning

EDUCATION

BRAC University <i>Bachelor of Science in Computer Science and Engineering</i>	Dhaka, Bangladesh 2018 – 2023
SFX Greenherald School <i>A Levels (Cambridge International Examination)</i>	Dhaka, Bangladesh 2017
Sunnydale School <i>O Levels (Cambridge International Examination)</i>	Dhaka, Bangladesh 2015

TECHNICAL SKILLS

Programming: Python, JavaScript, Java, PHP, R
Machine Learning: YOLO, OpenVINO, PyTorch
Data & Systems: Pandas, NumPy, PostgreSQL, MySQL, Redis
Backend: Django, Django REST Framework, FastAPI
Frontend: React, Svelte
Tools: Docker, Nginx, GitHub Actions, Git, Linux

RESEARCH EXPERIENCE

Fish Detection and Dataset Development (Ongoing) <i>Python, YOLO, Computer Vision</i> – Collecting and curating a real-world fish image dataset for detection tasks. – Building object detection models for species identification using YOLO. – Designing annotation and preprocessing pipelines for dataset scalability. – Focused on real-world deployment challenges such as noise, occlusion, and class imbalance. – Ongoing work with planned research publication.	Mar 2026 – Present
AlterEye: Vision-Based Factory Analytics <i>Python, OpenVINO, YOLO, GStreamer, MQTT, Redis</i> – Worked on a real time multi camera vision system for industrial monitoring and anomaly detection. – Developed pipelines for object detection and activity recognition using YOLO-based models. – Processed multi camera RTSP streams with GStreamer for scalable ingestion. – Used MQTT for lightweight, real-time data transfer between system components. – Stored intermediate results in Redis and persisted structured data in the database. – Designed end-to-end flow for low latency analytics and monitoring dashboards.	
Banknote Quality Classification for CDM Template Calibration <i>Document Analysis, Dataset Curation</i> – Built a banknote data collection pipeline at Technomedia Limited with NCR Corporation for CDM recognition training. – Categorized 12+ condition classes (ATM Fit, Counterfeit, Mutilated, Graffiti) on NCR SR61 and SS83 platforms. – Applied 4 orientation protocol (Face A/B/C/D) for feed alignment validation and rejection rate analysis. – Cross referenced output classifications (4A, 4B, CAT1–CAT3) from DAT files against physical samples. – Formalized structured datasets in Excel for NCR submission, improving counterfeit detection templates.	

PROFESSIONAL EXPERIENCE

Software Engineer

AlterSense Limited

Mar 2024 – Present

Dhaka, Bangladesh

- Automated 10+ industrial workflows using Python, reducing manual effort by 100+ hours/month and improving accuracy by 40%.
- Designed and developed scalable REST APIs in Django with optimized queries, caching, and robust testing.
- Analyzed real world operational data to identify inefficiencies and deliver targeted automation solutions.
- Deployed containerized services using Docker and Nginx, and implemented CI/CD pipelines via GitHub Actions.
- Mentored junior engineers in system design and development best practices.

Technical Support Engineer Intern

Technomedia Limited

May 2023 – Oct 2023

Motijheel, Dhaka, Bangladesh

- Diagnosed and resolved 50+ ATM hardware/software issues across field deployments, improving uptime by 30%.
- Collaborated with the Counterfeiting Department on hardware validation, reducing ATM failure rates by 25%.
- Contributed to a CDM banknote classification pipeline covering 12+ condition classes with a 4 orientation feed.
- Verified DAT classifications with physical samples and optimized NCR datasets, improving detection accuracy.

Student Tutor (O & A Levels)

Self-employed

2018 – 2023

Dhaka, Bangladesh

- Mentored 15+ students under the Cambridge curriculum, with over 90% achieving A grades.
- Focused on developing analytical thinking and structured problem-solving skills.

PROFESSIONAL PROJECTS

AlterBot | *Python, Django, Playwright, Selenium, Docker, Svelte, PostgreSQL*

- Developed automation systems for garment industry workflows, including data extraction, booking, and export monitoring.
- Reduced manual labor by 66%, saving 100+ hours weekly while improving system reliability.
- Built a real time monitoring dashboard and contributed to scalable backend architecture.

AlterEye | *Python, GStreamer, YOLO, MQTT, Redis*

- Processed multi camera RTSP streams using GStreamer for scalable ingestion.
- Built YOLO based pipelines for real time detection and monitoring.
- Designed scalable data flow using MQTT and Redis for low latency systems.

Technomedia Call Management System (TCMS) | *PHP, MySQL, JavaScript, HTML/CSS*

- Contributed to a full-stack web system for tracking ATM service requests, engineer assignments, and call logs.
- Developed frontend UI components and dashboards for real time service request monitoring and engineer wise reporting.
- Designed a normalized relational database schema (ER and class diagrams) to support multi entity call management.

VOLUNTEER AND EXTRACURRICULAR ACTIVITIES

JAAGO Foundation | *Volunteer*

2018

- Participated in educational fundraising campaigns supporting underprivileged children across Bangladesh.

Scouting | *Team Member*

2012 – 2015

- Represented school at the 13th Regional Scout Rally; built teamwork and leadership through community service activities.